Climate Data Exploration: Air Quality Analysis

1. Dataset Overview

The dataset contains air-quality observations with five selected fields: City, Date, PM2.5, PM10, and AQI (Air Quality Index). The cleaned sheet contains thousands of records, making it suitable for basic exploration of air-quality values.

2. Data Cleaning Process

I copied the five relevant columns into a separate sheet named Cleaned_Data and used Google Sheets' Remove duplicates feature to remove duplicate records. I checked the AQI column for blank values and sorted AQI from A to Z to help identify the minimum and maximum values. These steps made the data easier to inspect.

3. Key Patterns and Trends

The calculated average AQI is approximately 251.11. The maximum AQI in the inspected data is 500, while the minimum is 0. Both the highest and lowest values were observed in records for Delhi. The wide range from 0 to 500 shows substantial variation in the recorded AQI values. An average around 251 suggests that the dataset includes many high-AQI observations; the exact interpretation depends on the AQI scale used by the data source.

4. Anomalies or Unusual Observations

The AQI value of 500 is at the top of the observed range and deserves attention because it may represent an extreme pollution event or a capped reporting value. The AQI value of 0 is also worth checking against the original record and source documentation to confirm that it represents a valid measurement rather than a placeholder or missing-data code. These values should be verified before drawing firm conclusions.

5. Conclusion

This exercise helped me practice selecting relevant columns, removing duplicate records, checking for blanks, sorting data, and calculating summary statistics. A useful next step would be to compare average AQI across cities or dates and verify unusual readings using the dataset's source notes.